Overview of hardware components

Q-1. Draw a labeled diagram of a desktop PC showing the motherboard, CPU, RAM, GPU, storage (HDD/SSD), power supply, and cooling system. For each component, write its main function next to it.

Ans:



Q-2 Create a table listing each major PC component (motherboard, CPU, RAM, GPU, storage, power supply, cooling system), its typical location inside the case, and one example of how it connects or interacts with another component.

Hint: For example, "CPU sits in the motherboard socket and communicates with RAM via the memory slots."

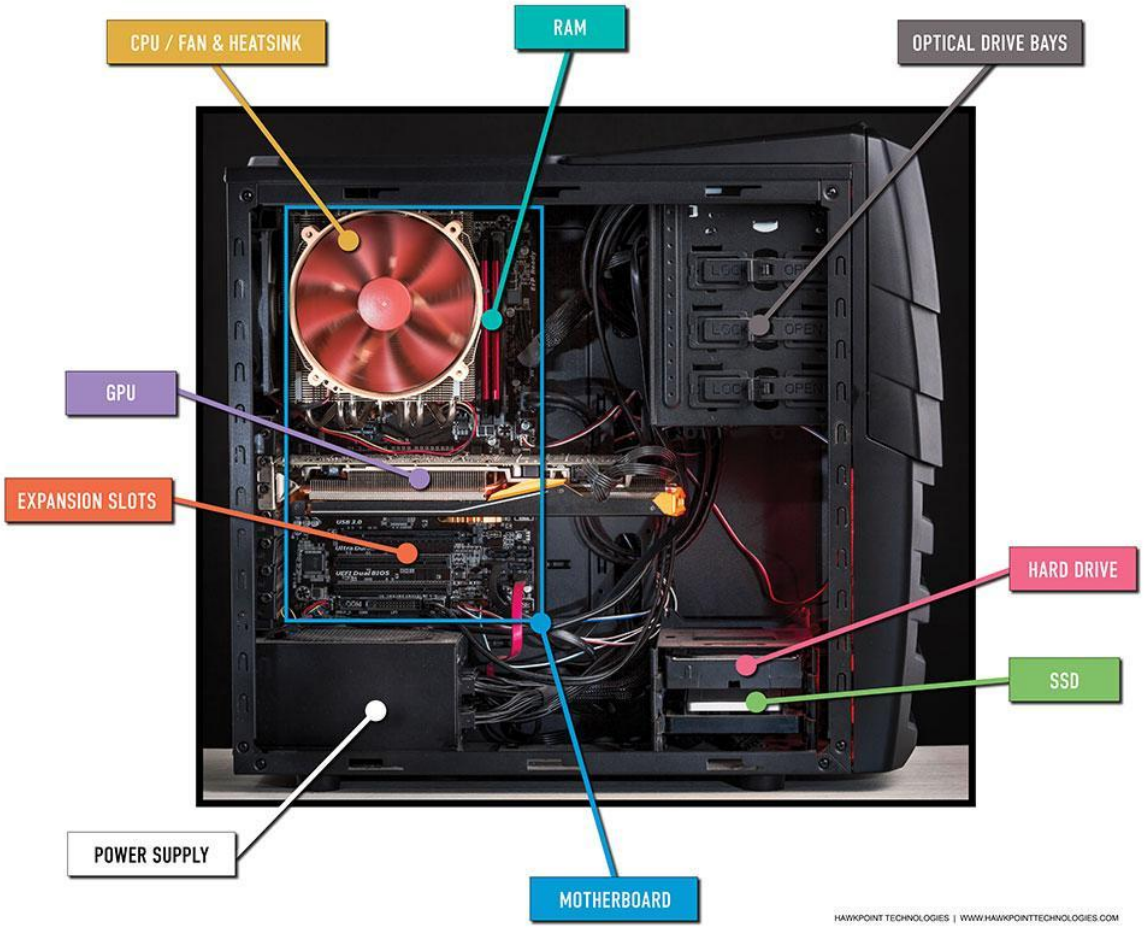
Ans: Major PC Components and Their Connections

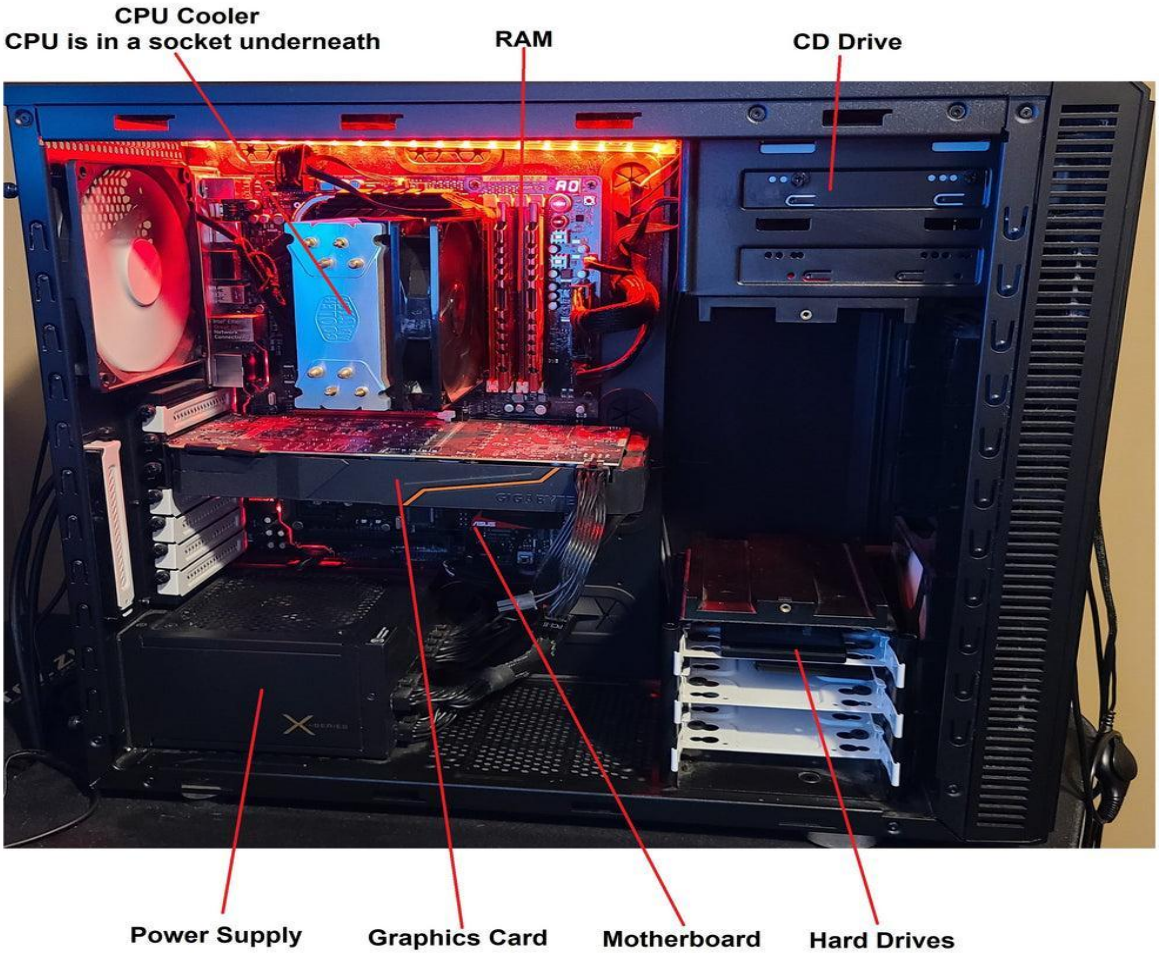
PC Component	**Typical Location Inside the Case**	**Example of Connection / Interaction**
Motherboard	Mounted inside the main body of the case	CPU, RAM, GPU, and storage devices connect to the motherboard.
CPU	Installed in the CPU socket on the motherboard	CPU sits in the motherboard socket and communicates with RAM through the memory slots.
RAM	Installed in RAM slots near the CPU on the motherboard	RAM connects to the motherboard and provides temporary data to the CPU.
GPU	Installed in a PCIe slot on the motherboard	GPU connects to the motherboard through the PCIe slot and processes graphics.
Storage (HDD/SSD)	Usually in storage bays or mounted on the case	SSD/HDD connects to the motherboard using SATA or M.2 and stores files and programs.
Power Supply (PSU)	Usually at the bottom or top of the case	PSU supplies power to the motherboard, CPU, GPU, and storage devices.
Cooling System	Around the CPU or mounted at the front/rear/top of the case	CPU cooler removes heat from the CPU, while case fans move hot air out.

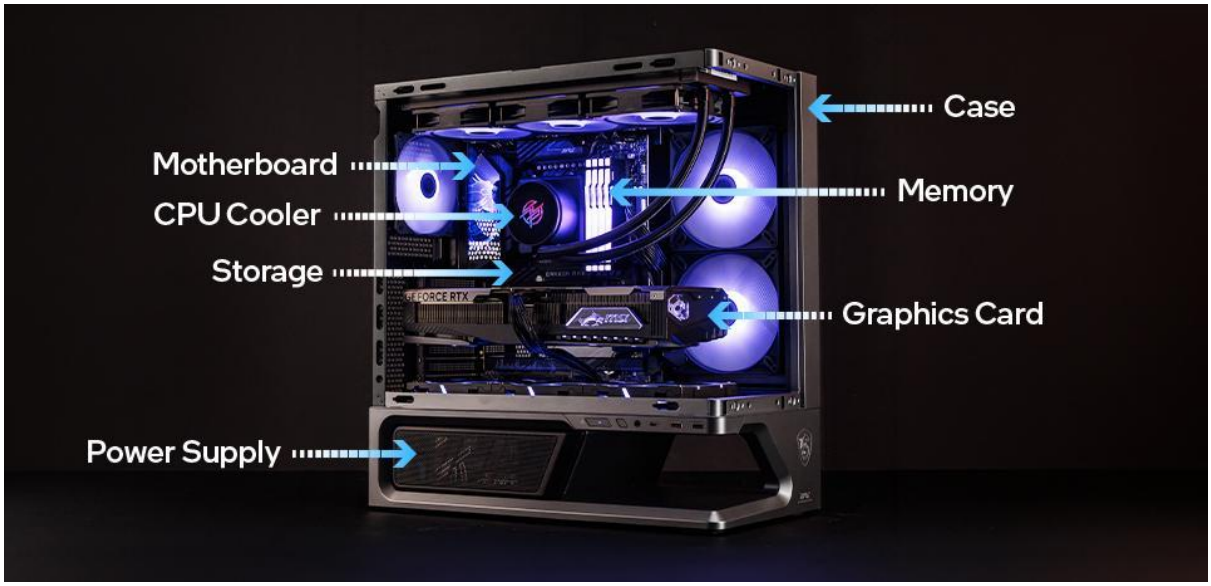
Q-3 Take a clear photo or screenshot of the inside of any desktop PC (your own, a friend's, or an image from the web) and use any image editor to digitally label at least 6 major components as discussed in class.

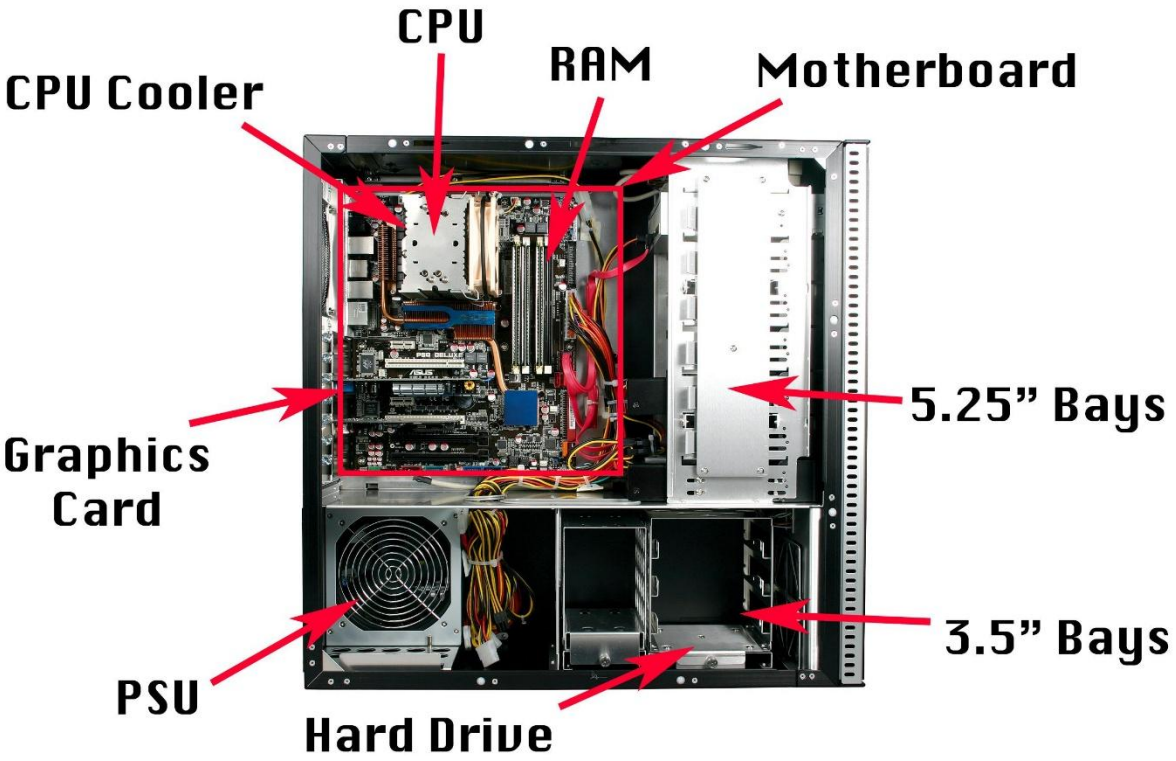
Ans: **3. Label the Inside of a Desktop PC**

For this task, you can use a clear desktop-PC image from the web and label the components using **MS Paint, Canva, PowerPoint, or any image editor**.

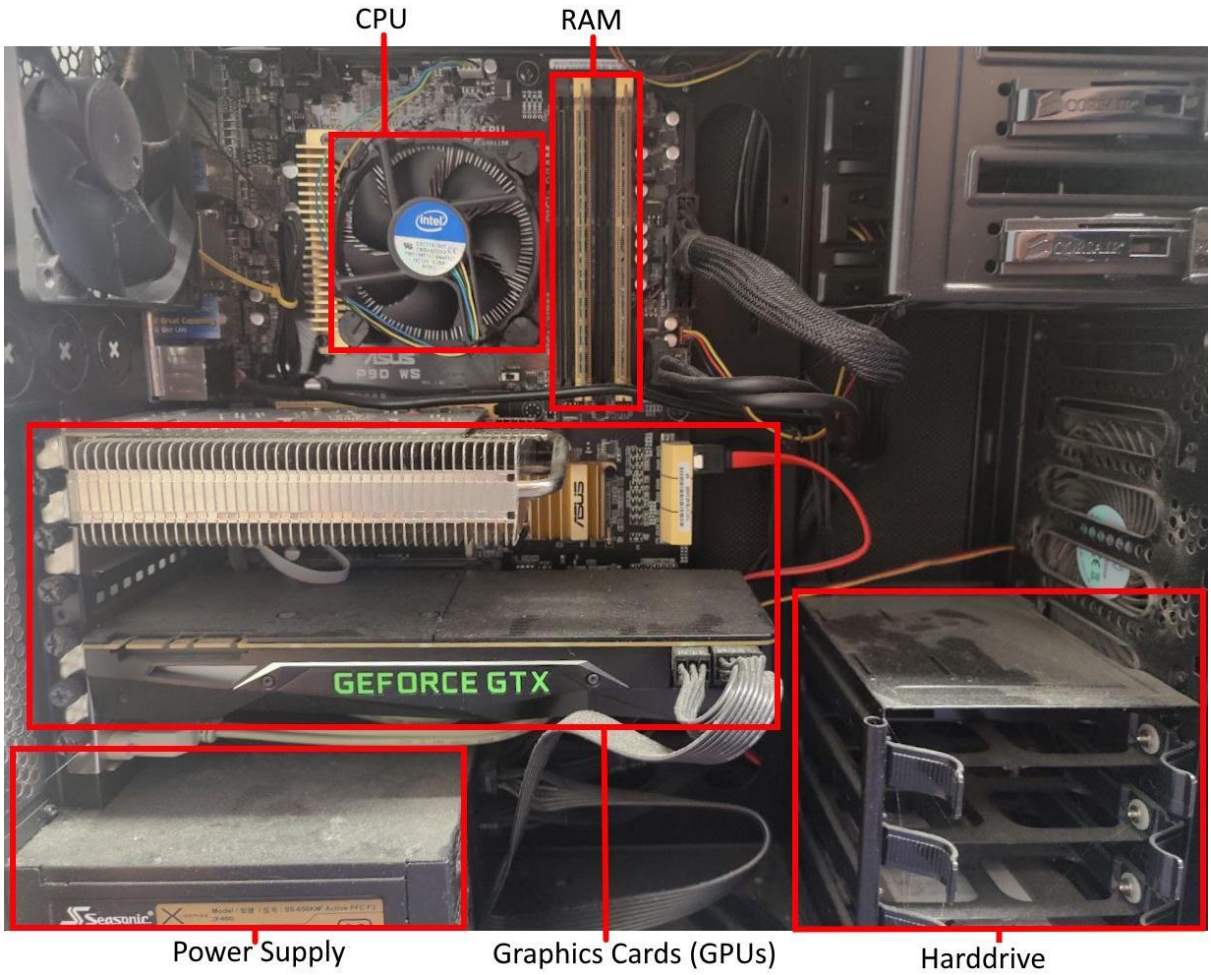



CPU Cooler
CPU is in a socket underneath
RAM
CD Drive
Power Supply
Graphics Card
Motherboard
Hard Drives


Case
Motherboard
Memory
CPU Cooler
Storage
Graphics Card
Power Supply


CPU
CPU Cooler
RAM
Motherboard
Graphics Card
5.25” Bays
3.5” Bays
PSU
Hard Drive


1 CPU (PROCESSOR)
The brain of the computer. Installs into the CPU socket on the motherboard.
2 CPU COOLER
Keeps the CPU cool. Connects to the motherboard via the CPU_FAN header.
3 RAM (MEMORY)
Stores data the CPU uses quickly. Installs in DIMM slots.
4 GPU (GRAPHICS CARD)
Handles graphics rendering. Installs in the PCIe x16 slot. May need power from PSU.
5 STORAGE (SSD / HDD)
Stores your OS, apps, and files. Connects via SATA cables or M.2 slot.
6 POWER SUPPLY (PSU)
Provides power to all components. Connects via motherboard 24-pin and 8-pin cables, plus others.
7 MOTHERBOARD
The central hub that connects everything together.
8 CASE FANS
Maintain airflow and cooling. Connect to SYS_FAN headers.
9 FRONT PANEL CONNECTORS
Power switch, reset switch, LEDs, USB, audio ports. Connect to motherboard front panel headers.



A suitable example image clearly shows the internal components such as the CPU/cooler, RAM, GPU, motherboard, power supply, and storage.

Labels to Add

Put arrows and these labels on the image:

1. **Motherboard** – Main circuit board connecting all components.
2. **CPU** – Processes instructions and data.
3. **RAM** – Temporary memory used while programs are running.
4. **GPU / Graphics Card** – Handles graphics and video.
5. **Storage (HDD/SSD)** – Stores files, programs, and the operating system.
6. **Power Supply (PSU)** – Supplies electrical power to the components.
7. **Cooling Fan / CPU Cooler** – Removes heat and keeps the PC cool.

Q-4 Write a short explanation (3-4 sentences) describing the data flow when you open a game on your PC, mentioning the roles of CPU, RAM, storage, GPU, and motherboard.

Hint: Start from storage loading data, then CPU processing, RAM usage, and GPU rendering.

Ans: **Data Flow When Opening a Game**

When you open a game, the **storage (HDD/SSD)** loads the game files and sends them to the **RAM** through the **motherboard**. The **CPU** processes the game instructions and uses the RAM to access the required data quickly. The **GPU** receives graphics data and renders the game images on the screen. The **motherboard** connects the CPU, RAM, GPU, and storage so they can communicate with each other.